Introducción a la Electrónica

Unidad 7

El Amplificador Operacional de Tensión (AOV)

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Parte 2

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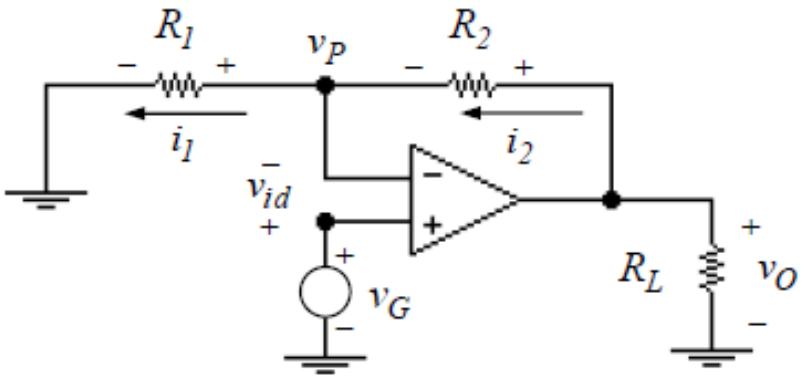
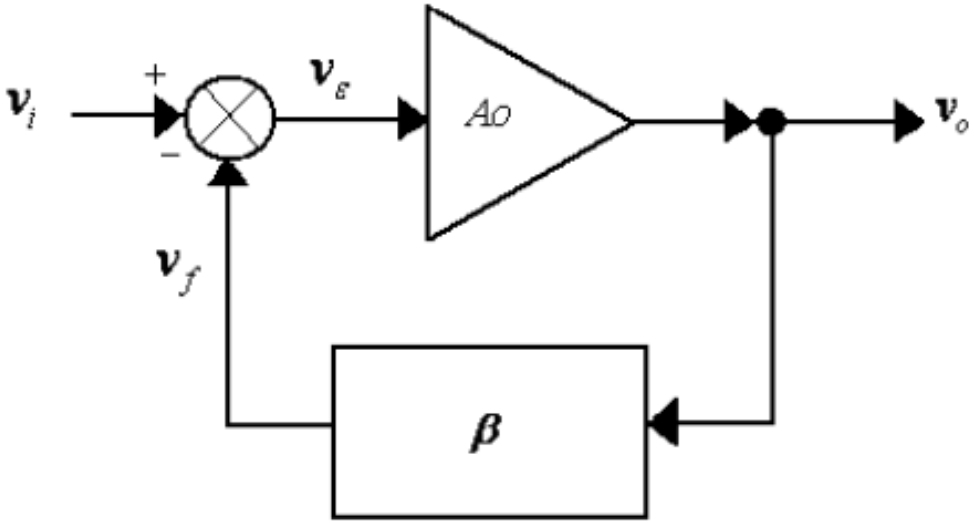
7.2 Amplificador Realimentado NO INVERSOR

Circuito



Sensa la tensión de salida y realimenta tensión

Diagrama en Bloques

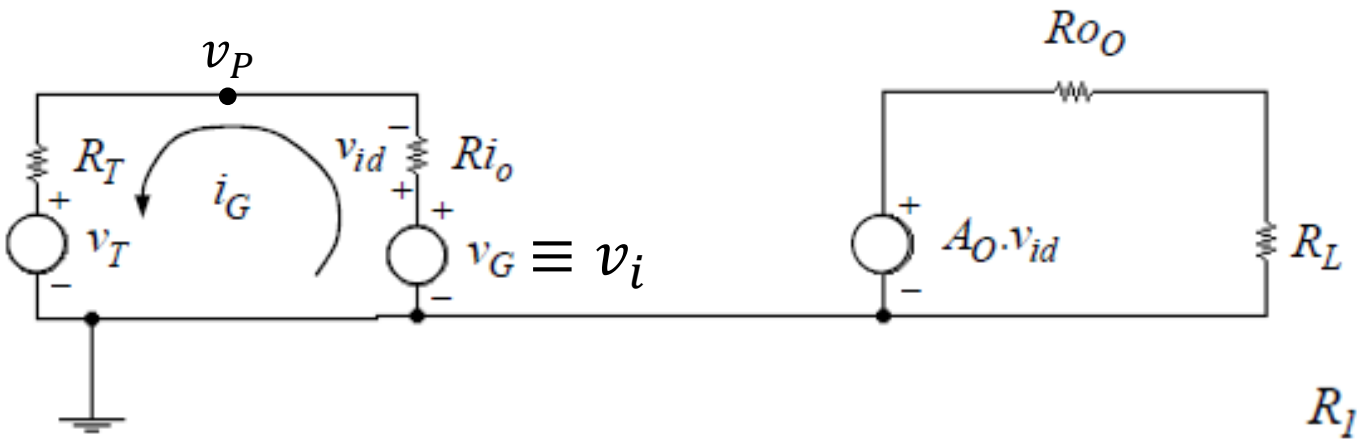


Modelado por Cuadripolo

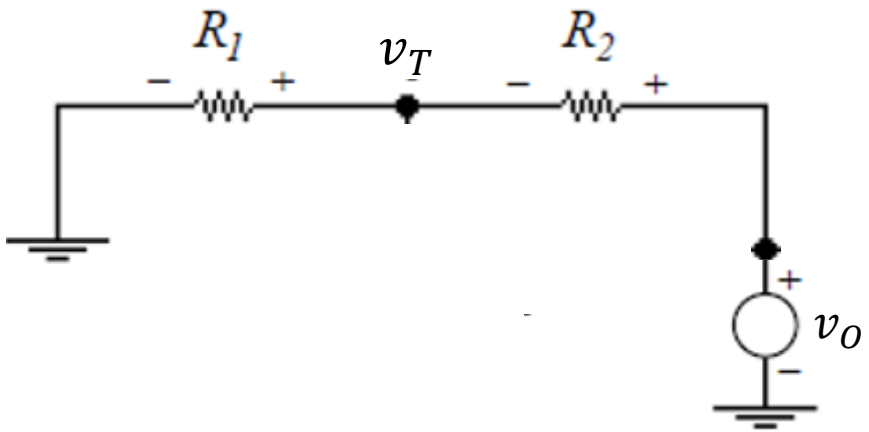
Sensa tensión de salida (v_o) y realimenta tensión (v_f)

$$R_T = R_1 // R_2$$

$$v_T = \frac{v_O}{(R_1 + R_2)} . R_1$$



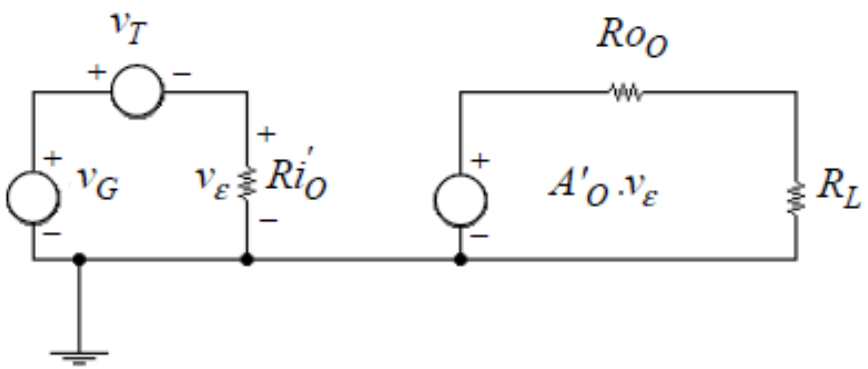
$$v_f \equiv v_T = v_o\left(\frac{R_1}{R_1 + R_2}\right) = v_o.\beta \quad \Rightarrow$$



$$\Rightarrow \quad \beta = \frac{R_1}{R_1 + R_2}$$

$$v_\varepsilon = v_i - v_f = \frac{v_{id}}{R_{i_O}} . (R_{i_O} + R_T) > 0 \quad \Rightarrow \quad v_{id} > 0$$

Análisis Detallado



$$Ao.v_{id} = \left(Ao.\frac{R_{i_O}}{R_{i_O} + R_T} \right).v_\varepsilon = A'_O.v_\varepsilon$$

$$v_o = \left(\frac{A'_O.R_L}{R_{O_O} + R_L} \right).v_\varepsilon = \left[\frac{Ao.R_{i_O}R_L}{(R_{i_O} + R_T).(R_{O_O} + R_L)} \right].v_\varepsilon = A''_O.v_\varepsilon$$

$$A''_O.\beta >> 1$$

$$\boxed{\frac{v_O}{v_G} = \frac{A''_O}{(1 + A''_O.\beta)} \cong \frac{1}{\beta} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}}$$

$$si\ el\ AOV\ es\ ideal\ \Rightarrow$$

$$\Rightarrow\ A''_O = A'_O = A_O$$

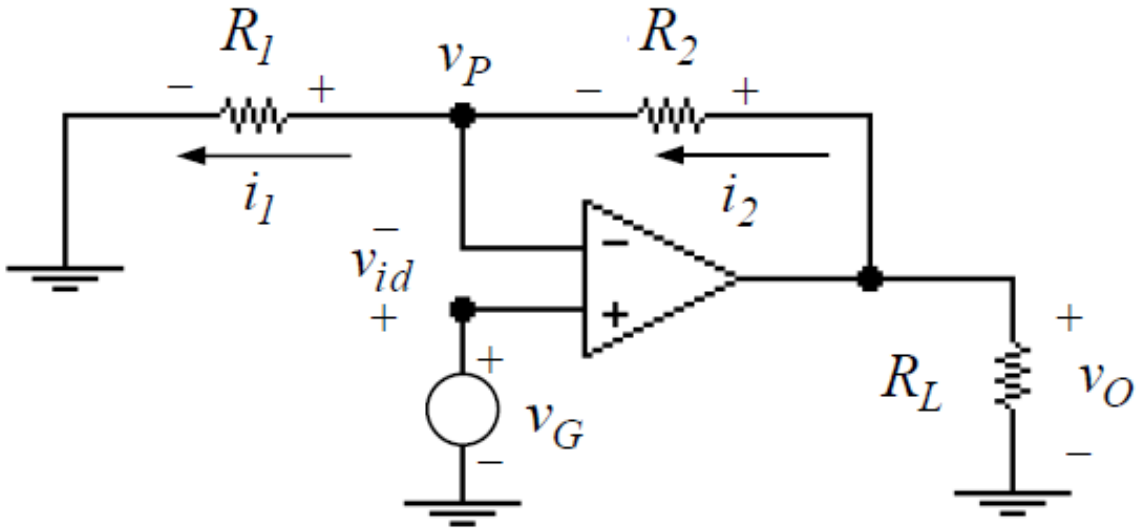
$$si \begin{cases} R_{i_O} \gg R_T \\ R_{O_O} \ll R_L \end{cases}$$

$$A''_O \cong A'_O \cong A_O$$

Análisis Simplificado

$$v_{id} \rightarrow 0 \ (MASA\ VIRTUAL)$$

$$v_p \cong v_G \ (Nuevamente:\ Tensión\ CERO,\ pero,\ SIN\ corriente)$$



$$i_G = \frac{v_{id}}{R_{io}} \rightarrow \frac{0}{\infty} \therefore i_G \rightarrow 0$$

$$i_1 = \frac{v_P}{R_1} = \frac{v_G}{R_1}$$

$$i_1 = i_2$$

$$\frac{v_G}{R_1} = \frac{v_O - v_G}{R_2} \Rightarrow \frac{v_O}{R_2} = v_G\left(\frac{1}{R_2} + \frac{1}{R_1}\right)$$

$$\boxed{\frac{v_O}{v_G} = \left(1 + \frac{R_2}{R_1}\right)}$$